COURSE DESCRIPTION			
Program	**Food Processing Technology**		
Course Title	**Fundamentals of Food Processing Technology**		
Course Code	**1421**	**Semester**	**I**
Type of Course	**Practicum**	**Contact Hours**	**90**
Teaching Scheme (L: T:P:R)	**3:0:3:0**	**Credits**	**4.5**
CIE Marks	**40**	**ESE Marks**	**100**

Introduction

Fundamentals of Food Processing Technology is an introductory core subject for first-semester Diploma students that provides basic knowledge of food components and their role in food processing and preservation. The subject covers fundamental concepts related to macro and micronutrients, water in food systems, carbohydrates, proteins, lipids, vitamins, and minerals, which form the foundation for advanced studies in food processing and quality control. Designed as a practicum, the course integrates theory with practical sessions, enabling students to apply theoretical knowledge through laboratory experiments, food product preparation, and basic analytical techniques. This integrated approach helps students understand the relationship between food composition, processing, quality, and consumer acceptability while developing essential laboratory skills required in the food industry.

Course Objectives

1	To impart fundamental knowledge on the composition, classification, and functional role of major food constituents.
2	To develop understanding of the properties of water and its significance in food processing and preservation.
3	To provide hands-on experience in basic food processing operations and preparation of selected food products.
4	To develop basic laboratory skills for analysis and evaluation of food quality through simple analytical and sensory methods.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge of chemistry concepts such as acids, bases, and solutions			Secondary Education
Awareness of common food materials, nutrients, and basic laboratory safety practices.			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the composition, classification, and functional role of major food constituents	Understand
CO2	Describe the properties of water and its significance in food processing and preservation.	Understand
CO3	Perform basic food processing operations and preparation of selected food products.	Apply
CO4	Apply simple analytical and sensory evaluation techniques to assess food quality.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	1
CO2	3	2	-	-	-	1	-	-	-	-	1
CO3	2	2	3	2	2	1	-	1	-	-	-
CO4	2	3	-	3	2	-	-	-	1	-	1
Total	10	9	3	5	4	2	-	1	1	-	3
Correlation Level	2.5	2.25	3	1.67	2	1	-	1	1	-	1
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	

3	0	3	0	6	90	4.5	20	60	80	20	40	60	140
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Explain the form and properties of water in food	List the different macro and micro nutrient present in food. Describe the Water activity and its importance in food. Illustrate the different form, structure and properties of water in food.	10	0
Module I	Fundamentals of Water and Nutrients in Food Systems	Study of food processing laboratory guidelines, safety measures, and basic equipment. Determination of moisture content of selected food materials by oven drying method. Determination or demonstration of water activity using standard food samples. Study the effect of blanching on selected vegetables (colour, texture, moisture loss)	0	11
Module II	Illustrate the composition and classification of carbohydrates	Describe the chemistry of carbohydrates. Outline the composition and structure of carbohydrates. Explain the classification and properties of carbohydrates. Illustrate various sources and functional importance of carbohydrates	12	0
Module II	Chemistry, Properties, and Functional Role of Carbohydrates	Detection of carbohydrates in selected food samples using standard qualitative tests. Estimation of total carbohydrate content in a given food sample by a suitable method. Preparation of jam using fruit pulp and sugar. Preparation of toffee using sugar, milk solids, and fat.	0	12
Module III	Describe the structure and categories of protein in food	Describe the functional role of protein. Explain the structure of protein. Illustrate the type of classification of protein.	12	0
Module III	Structure, Classification, and Functional Role of Proteins in Food	Detection of proteins in selected food samples using standard qualitative tests. Estimation of protein content in a given food sample by a suitable method. Preparation of paneer from milk by acid coagulation.. Preparation of mayonnaise and study of emulsion stability.	0	6
Module IV	Explain the importance of micro nutrients and lipids	Explain the classification and properties of lipids. Illustrate the types and sources of vitamins. Describe the types and functional role of minerals.	9	0
Module IV	Lipids, Vitamins, and Minerals: Properties and Functional Roles in Food	Detection of fats and oils in selected food samples using standard qualitative tests. Preparation of toffee and study of the role of fat in texture and mouthfeel. Preparation of tomato ketchup with emphasis on ingredient composition. Preparation of sauerkraut by natural fermentation.	0	6

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
List the different macro and micro nutrient present in food	Identify and classify the major macro and micronutrients present in food.	L	CO1	Understand	4
Describe the Water activity and its importance in food	Explain the significance of water activity in food quality and preservation	L	CO1	Understand	4
Illustrate the different form, structure and properties of water in food	Describe the forms, structure, and properties of water in food systems.	L	CO1	Understand	4

Identification and classification of food samples based on macro and micronutrient composition	Classify common food samples based on their major macro and micronutrient composition.	P	CO1	Apply	3
Qualitative identification of major food nutrients using simple food tests (starch, protein and fat)	Perform simple qualitative tests to identify major nutrients present in food samples.	P	CO1	Apply	3
Determination of water activity in selected food samples	Determine water activity of food samples and relate it to food quality and shelf life.	P	CO1	Apply	3
Determination of moisture content of food samples by oven drying method	Determine the moisture content of food samples and interpret the results	P	CO1	Apply	3
Module II					
Describe the chemistry of carbohydrates	Explain the basic chemistry and characteristics of carbohydrates. .	L	CO2	Understand	3
Outline the composition and structure of carbohydrates	Describe the composition and structural forms of carbohydrates.	L	CO2	Understand	3
Explain the classification and properties of carbohydrates	Classify carbohydrates based on their structure and properties.	L	CO2	Understand	3
Illustrate various sources and functional importance of carbohydrates	Explain the sources and functional role of carbohydrates in food processing.	L	CO2	Understand	3
Identification of carbohydrates using Molisch's test and Benedict's test	Perform qualitative tests to identify carbohydrates in food samples.	P	CO2	Apply	3
Identification of starch in different food samples using iodine test	Identify the presence of starch in different food samples using qualitative tests.	P	CO2	Apply	3
Determination of total soluble solids (TSS) of sugar solutions using a hand refractometer	Determine the total soluble solids of food samples using a refractometer.	P	CO2	Apply	3
Preparation of jam and evaluation of its quality	Prepare jam and relate the functional role of carbohydrates in product quality and preservation.	P	CO2	Apply	3
Module III					
Describe the functional role of protein	Explain the functional role of proteins in food processing.	L	CO3	Understand	3
Explain the structure of protein	Describe the structural organization of proteins	L	CO3	Understand	5
Illustrate the type of classification of protein	Classify proteins based on composition and structure.	L	CO3	Understand	4
Preparation of mayonnaise and study of emulsion stability.	Prepare mayonnaise and relate the emulsifying property of proteins to product stability.	P	CO3	Apply	3
Detection of proteins in selected food samples using standard qualitative tests.	Identify proteins in food samples using standard qualitative tests.	P	CO3	Apply	3
Estimation of protein content in a given food sample by a suitable method.	Estimate the protein content of food samples using standard analytical methods.	P	CO3	Apply	3
Preparation of paneer from milk by acid coagulation.	Prepare paneer by acid coagulation and relate protein denaturation and coagulation to product formation.	P	CO3	Apply	3
Module IV					
Explain the classification and properties of lipids	Explain the classification, properties, and functions of lipids in food.	L	CO4	Understand	3
Illustrate the types and sources of vitamins	Describe the types, sources, and importance of vitamins in foods.	L	CO4	Understand	3
Describe the types and functional role of minerals	Explain the types and functional role of minerals in human nutrition.	L	CO4	Understand	3
Detection of fats and oils in selected food samples using standard qualitative tests.	Identify fats and oils in food samples using qualitative tests.	P	CO4	Apply	3

Preparation of tomato ketchup with emphasis on ingredient composition.	Prepare tomato ketchup and relate the role of vitamins and ingredients in product quality.	P	CO4	Apply	3
Preparation of sauerkraut by natural fermentation.	Prepare sauerkraut by natural fermentation and explain its role in food preservation.	P	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	List five commonly consumed foods and classify them based on major macro and micro nutrients present.	6
2	Observe any packaged food label and identify information related to moisture or water content.	6
3	Prepare a short note on the importance of water activity in preventing food spoilage.	6
4	Observe blanching of any vegetable at home and record changes in colour and texture.	6
Module II		
5	Identify carbohydrate-rich foods commonly used at home and classify them based on their type.	6
6	Collect labels of three processed foods and note the form of carbohydrates mentioned.	6
7	Observe jam or jelly preparation at home and list the role of sugar in the product.	6
8	Prepare a brief chart showing sources and functional roles of carbohydrates in food.	6
Module III		
9	List protein-rich foods of plant and animal origin available locally.	6
10	Observe milk curdling during paneer or curd preparation and note the changes.	6
11	Collect food labels and identify protein content mentioned in nutritional information.	6
12	Prepare a simple note on different functional roles of proteins in food products.	6
Module IV		
13	Identify fat-rich foods used in daily cooking and note their physical characteristics.	6
14	List fruits and vegetables rich in different vitamins consumed at home.	6
15	Preparation for examination	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Food Processing Technology: Principles and Practice	P.J. Fellows	Woodhead Publishing
2	Fundamentals of Food Processing and Technology	W.A. Gould	Elsevier

REFERENCE			
Sl. No.	Title	Author	Publication
1	Textbook on Fundamentals of Food Technology	Jagbir Rehal, Jaspreet Kaur,	IIP Publishing
2	Fundamentals and Operations in Food Process Engineering	Susanta Kumar Das & Madhusweta Das	CRC Press

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.researchgate.net/publication/370842237_Food_Processing_Technology_Principles_and_Practice?utm_source=chatgpt.com	All
2	https://www.sciencedirect.com/book/9780128231296/introduction-to-food-engineering?utm_source=chatgpt.com	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Electric Oven	Laboratory oven with adjustable temperature 50–200 °C	1

2	Moisture Analyzer / Dish Set	Standard moisture dish (for drying moisture content)	1-2 sets
3	Sensory Evaluation Sheets	Standard forms	30–50 sheets

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40
Converted to			10	10	10	10	10		
Marks	5	10	10	15		10		60	40
Total	40							100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60